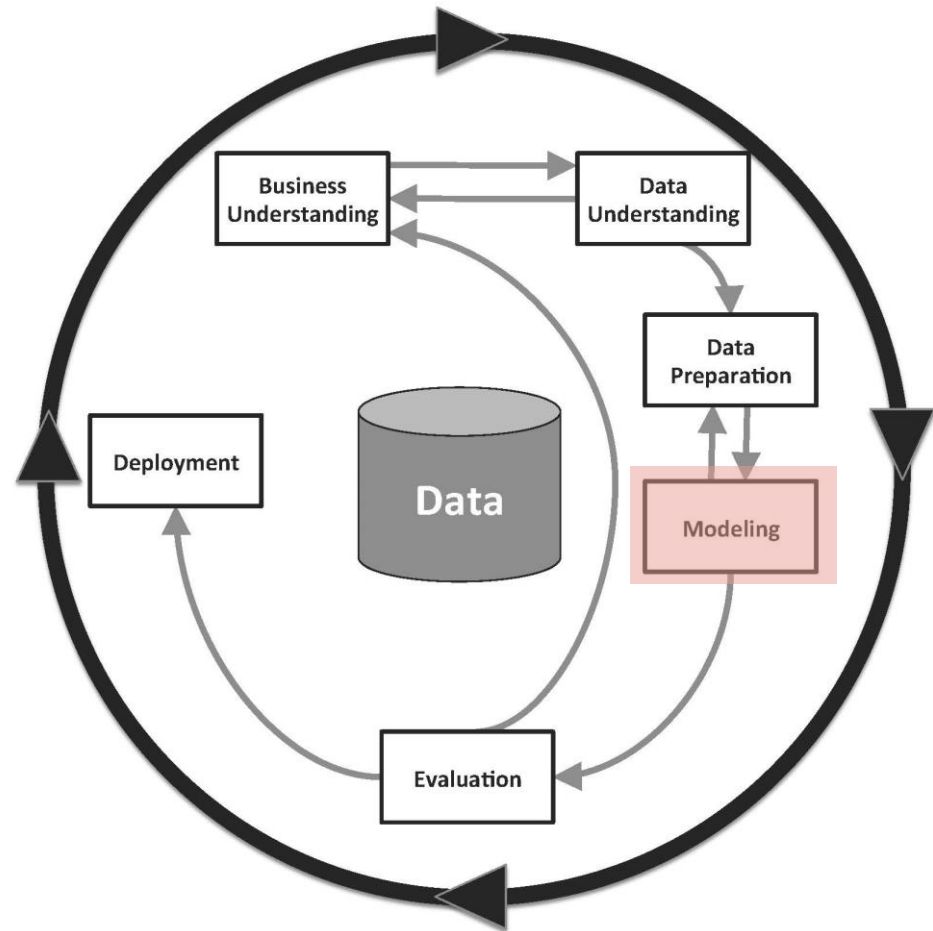


4. INFORMATION-BASED LEARNING (1)

SECTIONS 4.1~4.3



- **Information-based Learning**
- Similarity-based Learning
- Probability-based Learning
- Error-based Learning
- Deep Learning



Information is the resolution of uncertainty.

—Claude Elwood Shannon



信息论之父-香农

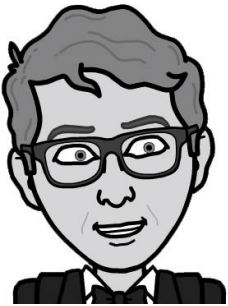
(1916年--2001年)

- 20岁就从密歇根大学数学系毕业
- 24岁获MIT数学博士学位
- 25岁加入贝尔实验室数学部
- 1948年发表论文 “A Mathematical Theory in Communication”
- 40岁重返MIT任数学教授，直至2001年逝世（Emeritus Prof. 荣休教授）

- In this chapter we are going to introduce a machine learning algorithm that tries to build predictive models **using only the most informative features**.
- In this context an **informative feature** is a **descriptive feature** whose values split the instances in the dataset into **homogeneous sets** with respect to the target feature value.

CONTENTS

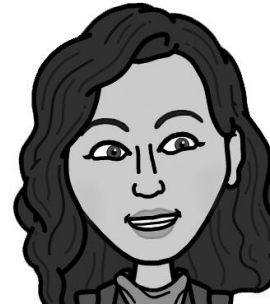
- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**



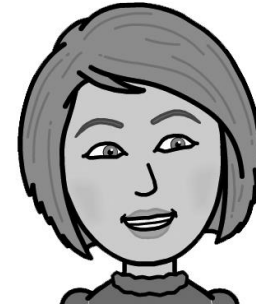
(a) Brian



(b) John



(c) Aphra



(d) Aoife

Figure 4.1: Cards showing character faces and names for the *Guess-Who* game

Man	Long Hair	Glasses	Name
Yes	No	Yes	Brian
Yes	No	No	John
No	Yes	No	Aphra
No	No	No	Aoife

Table 4.1: A dataset that represent the characters in the *Guess-Who* game

Which question would you ask first:

1. Is it a man?
2. Does the person wear glasses?
3. Does the person have long hair?

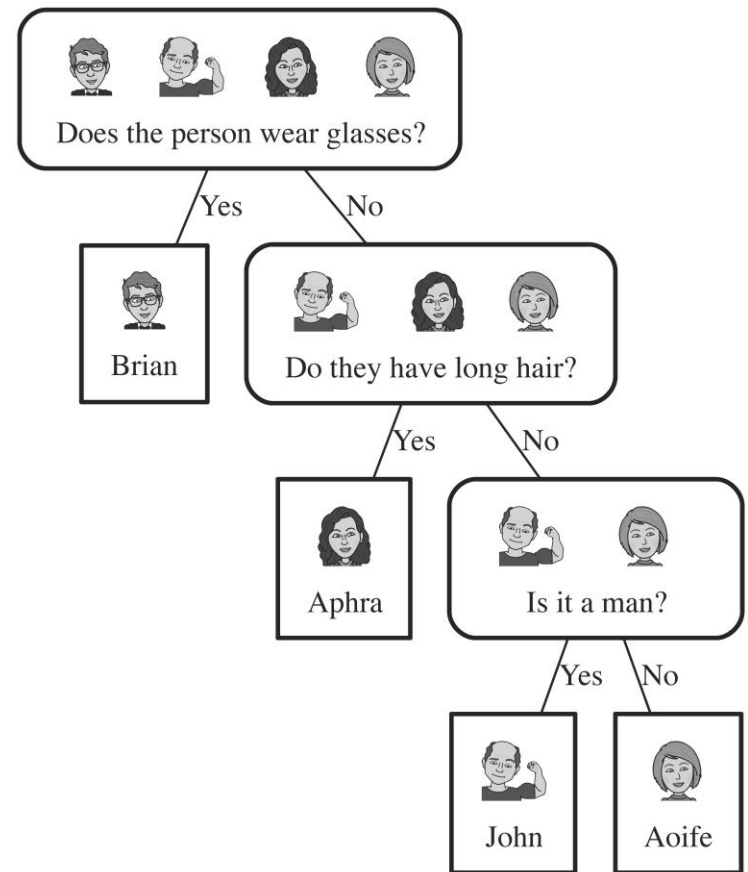
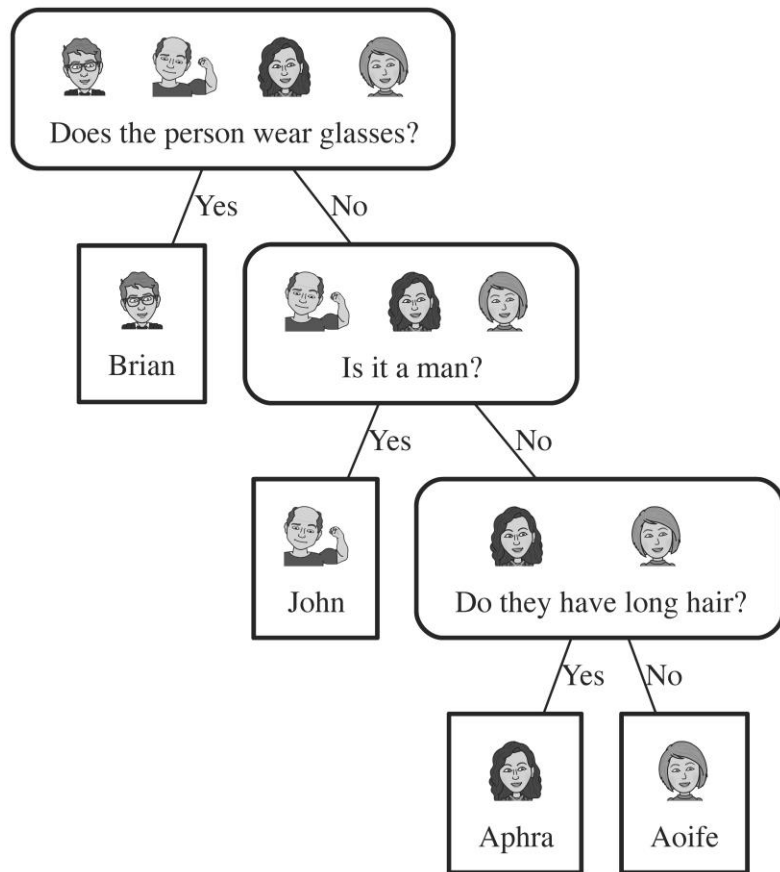


Figure 4.2: The different question sequences that can follow in a game of *Guess-Who* beginning with the question **Does the person wear glasses**

- In both of the diagrams:
 - one path is 1 question long
 - one path is 2 questions long
 - and two paths are 3 questions long
- Consequently, if you ask Question-2 (Does the person wear glasses?) first, the average number of questions you have to ask per game is:

$$\frac{1+2+3+3}{4} = 2.25$$

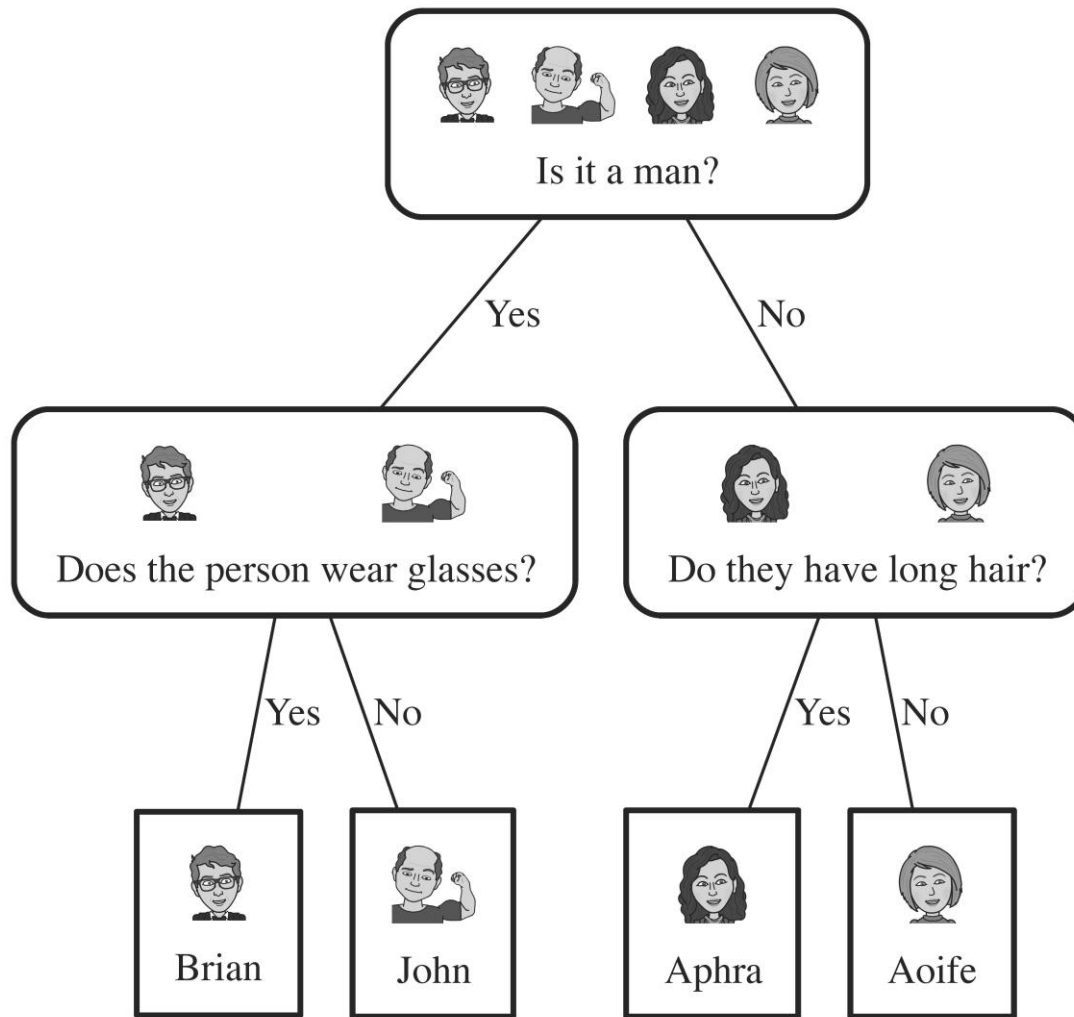
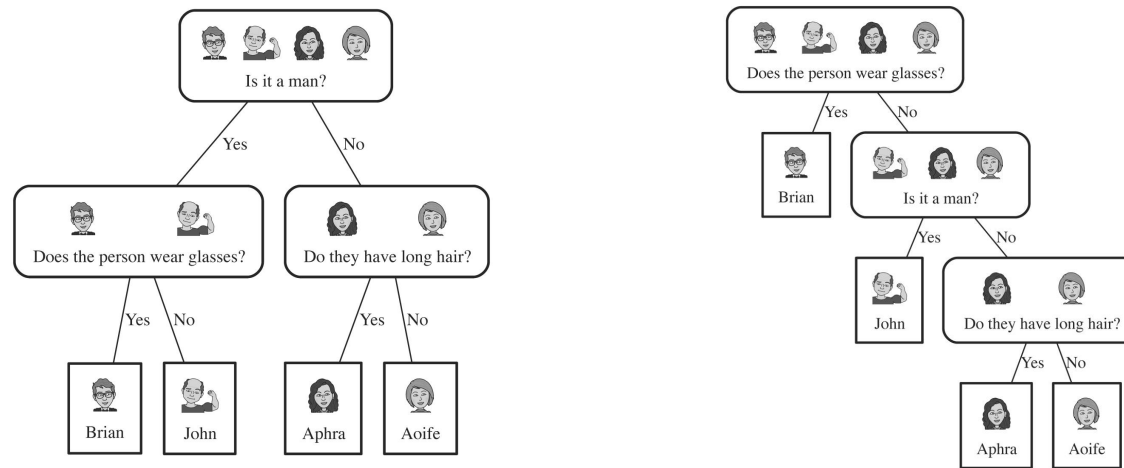


Figure 4.3: The different question sequences that can follow in a game of *Guess-Who* beginning with the question **Is it a man?**

- All the paths in this diagram are two questions long.
- On average, if you ask Question-1 (**Is it a man?**) first, the average number of questions you have to ask per game is:

$$\frac{2+2+2+2}{4} = 2$$



- On average getting an answer to Question-1 seems to give you **more information** than an answer to Question-2 or 3: less follow up questions.
- This is not because of the literal message of the answers: YES or NO.
- It is to do with how the answer to each questions **splits** the domain into different sized sets based on the value of the descriptive feature the question is asked about, and the **likelihood** of each possible answer to the question.

Big Idea

- So the big idea here is to figure out which features are the **most informative ones** to ask questions about by considering the effects of the different answers to the questions, in terms of:
 1. how the domain is split up after the answer is received
 2. the likelihood of each of the answers

CONTENTS

- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**

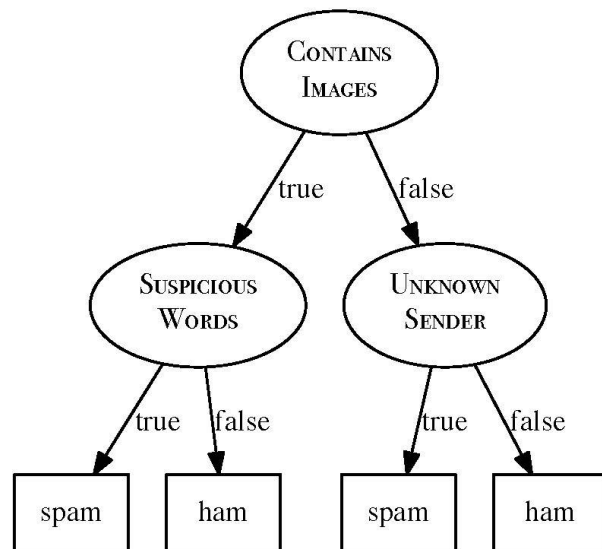
1. DECISION TREE

- A **decision tree** consists of:
 - a **root node** (or starting node)
 - **interior nodes**
 - **leaf nodes** (or terminating nodes)
- Each of the **non-leaf nodes** (root and interior) in the tree specifies a test to be carried out on one of the query's descriptive features.
- Each of the **leaf nodes** specifies a predicted classification for the query.

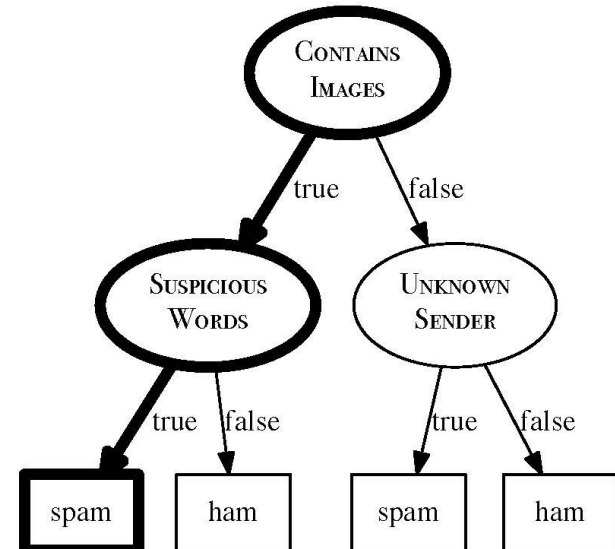
1. DECISION TREE

Table 4.2: An email spam prediction dataset.

ID	SUSPICIOUS WORDS	UNKNOWN SENDER	CONTAINS IMAGES	CLASS
376	true	false	true	spam
489	true	true	false	spam
541	true	true	false	spam
693	false	true	true	ham
782	false	false	false	ham
976	false	false	false	ham



(a)



(b)

Figure 4.4: (a) show a **decision tree** that is consistent with the instances in the spam dataset.

(b) shows the path taken through the tree shown in (a) to make a prediction for the query instance: **CONTAINS IMAGES = 'true'**, **SUSPICIOUS WORDS = 'true'**, **UNKNOWN SENDER = 'true'**

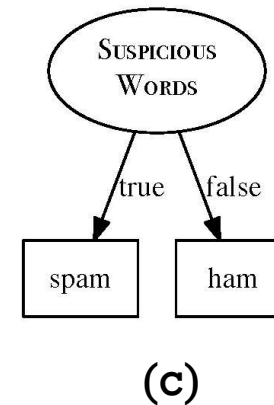
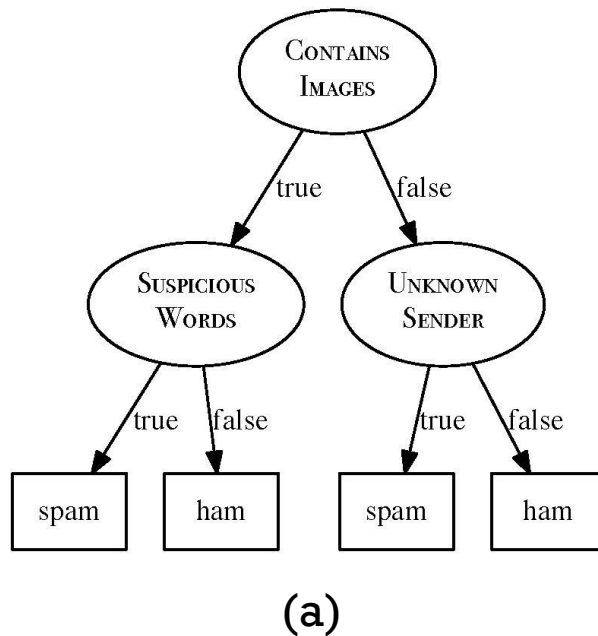


Figure 4.4: (a) and (c) show **two decision trees** that are consistent with the instances in the spam dataset.

1. DECISION TREE

- Both of these trees will return identical predictions for all the examples in the dataset.
- So, which tree should we use?

1. DECISION TREE

- Apply the same approach as we used in the *Guess-Who* game: prefer decision trees that use less tests (shallower trees).
- This is an example of **Occam's Razor**.
 - the principle of **Occam's Razor**: in explaining a thing, no more assumptions should be made than are necessary.

简约之法则

1. DECISION TREE

How do we create shallow trees?

- The tree that tests SUSPICIOUS WORDS at the root is very shallow because the SUSPICIOUS WORDS feature **perfectly** splits the data into pure groups of 'spam' and 'ham'.
- Descriptive features that split the dataset into **pure sets** with respect to the target feature provide information about the target feature.
- So we can make shallow trees by testing the informative features early on the tree.
- All we need to do that is a **computational metric** of the purity of a set: **entropy**

CONTENTS

- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**

2. SHANNON'S ENTROPY MODEL

- Claude Shannon's entropy model defines a computational measure of the **impurity** of the elements of a set.
- An easy way to understand the entropy of a set is to think in terms of the **uncertainty** associated with guessing the result if you were to make a random selection from the set.

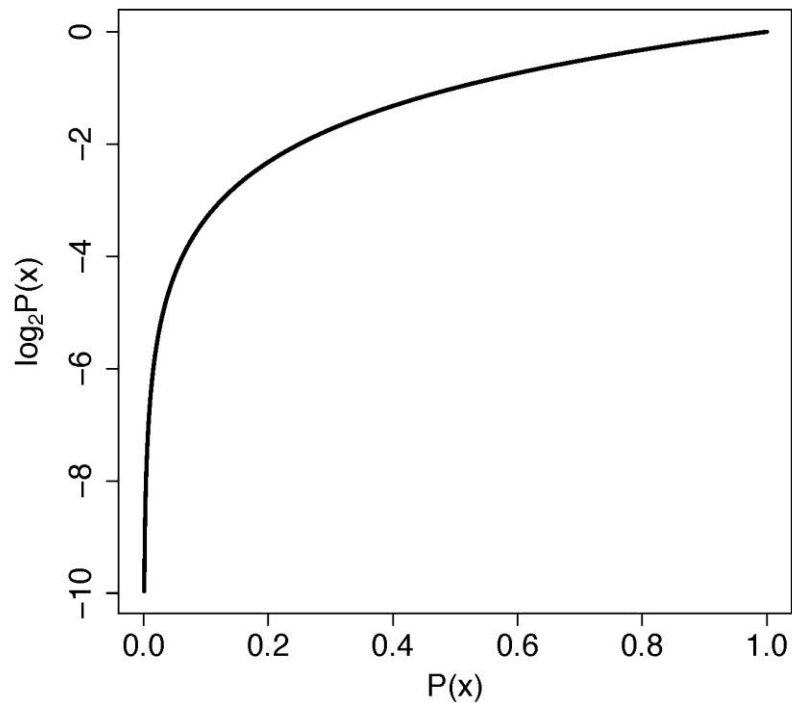
2. SHANNON'S ENTROPY MODEL

- Entropy is related to the probability of a outcome.
 - High probability ! Low entropy
 - Low probability ! High entropy

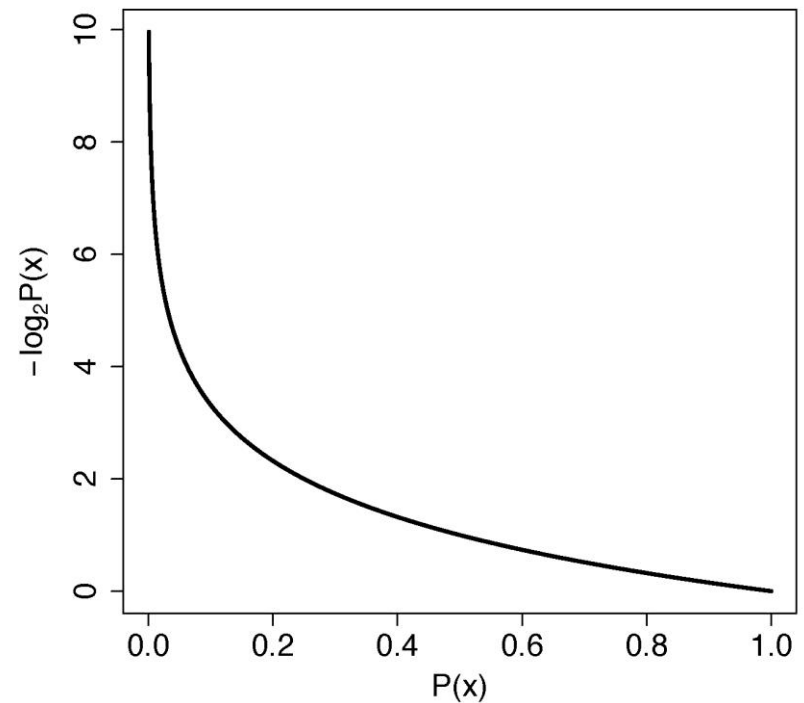
For example, suppose you visited an oracle who could answer any “yes or no” question. If you asked, “Will I live to be 125?”...

- R. J. McElice, The Theory of Information & Coding,
Cambridge University Press, 1978, 2002

- If we take the log of a probability and multiply it by -1 we get this mapping!



(a)



(b)

Figure 4.6: (a) A graph illustrating how the value of a binary log (the log to the base 2) of a probability changes across the range of probability values. (b) the impact of multiplying these values by -1.

2. SHANNON'S ENTROPY MODEL

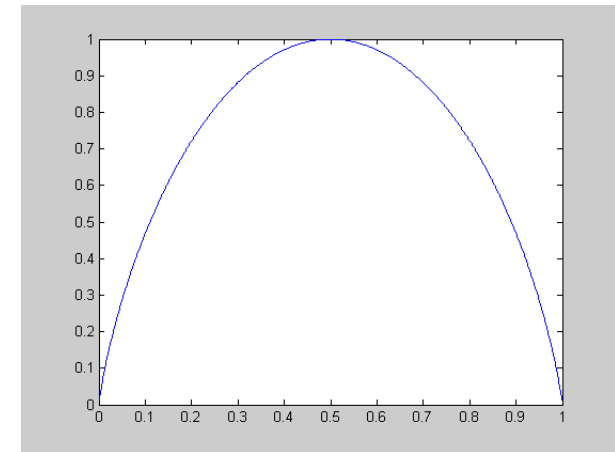
- Shannon's model of entropy is a **weighted sum** of the logs of the probabilities of each of the possible outcomes when we make a random selection from a set.

$$\begin{aligned} H(t) &= - \sum_{i=1}^l P(t=i) \times \log(P(t=i)) & (4.1) \\ &= \sum_{i=1}^l P(t=i) \times \log \frac{1}{P(t=i)} \end{aligned}$$

$$0 \leq H(t) \leq \log l$$

2. SHANNON'S ENTROPY MODEL

$$H(t) = - \sum_{i=1}^l P(t=i) \times \log(P(t=i))$$



$l=2$

- This definition needs elaboration:
 - The base of the logarithm is purposely left unspecified
 - If necessary, however, we shall denote the base- s and say that the entropy is being measured in base- s units.
 - Base-2 units are called *bits*, and base-e units are called *nats*.
 - If $P(t=i) = 0$, the term $P(t=i) \times \log P(t=i)$ is indeterminate, we define it to be 0.
 - If l is infinite, the sum may not converge, in this case we set $H(t) = +\infty$.

2. SHANNON'S ENTROPY MODEL

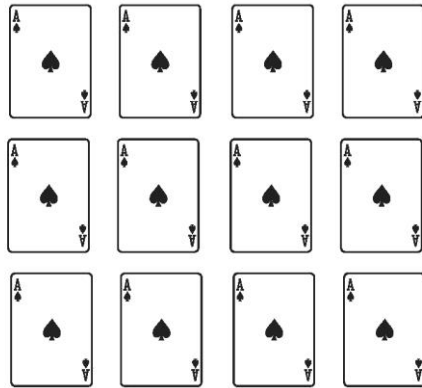
- What is the entropy of a set of 52 different playing cards?

$$\begin{aligned} H(card) &= - \sum_{i=1}^{52} P(card = i) \times \log_2(P(card = i)) \\ &= - \sum_{i=1}^{52} \frac{1}{52} \times \log_2\left(\frac{1}{52}\right) \\ &= 5.700 \text{ bits} \end{aligned}$$

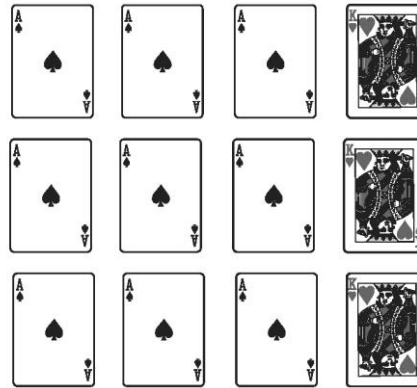
2. SHANNON'S ENTROPY MODEL

- What is the entropy of a set of 52 playing cards if we only distinguish between the cards based on their suit $\{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}$?

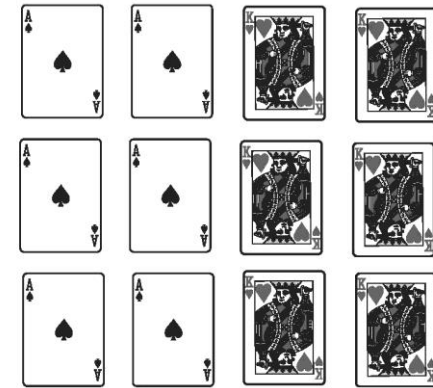
$$\begin{aligned} H(\text{suit}) &= - \sum_{l \in \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\}} P(\text{suit} = l) \times \log_2(P(\text{suit} = l)) \\ &= - \left[P(\heartsuit) \times \log_2(P(\heartsuit)) + P(\clubsuit) \times \log_2(P(\clubsuit)) \right. \\ &\quad \left. + P(\diamondsuit) \times \log_2(P(\diamondsuit)) + P(\spadesuit) \times \log_2(P(\spadesuit)) \right] \\ &= - \left[\frac{13}{52} \times \log_2\left(\frac{13}{52}\right) + \frac{13}{52} \times \log_2\left(\frac{13}{52}\right) \right. \\ &\quad \left. + \frac{13}{52} \times \log_2\left(\frac{13}{52}\right) + \frac{13}{52} \times \log_2\left(\frac{13}{52}\right) \right] \\ &= 2\text{bits} \end{aligned}$$



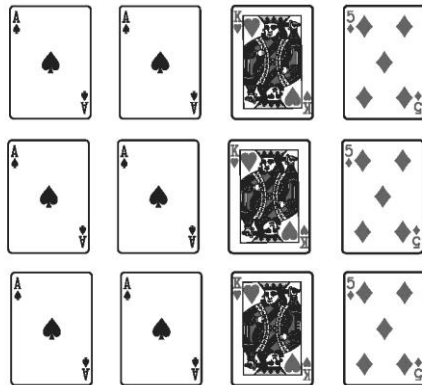
(a) $H(\text{card}) = 0.00$



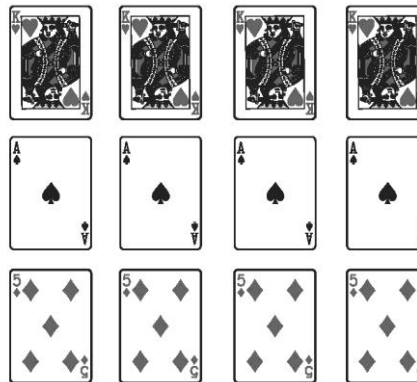
(b) $H(\text{card}) = 0.81$



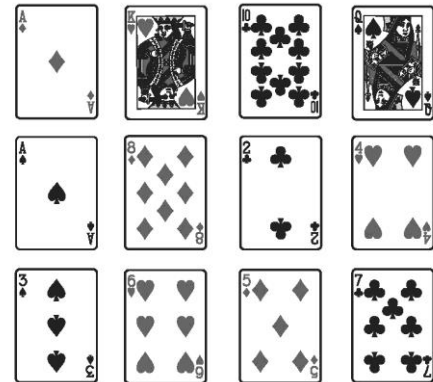
(c) $H(\text{card}) = 1.00$



(d) $H(\text{card}) = 1.50$



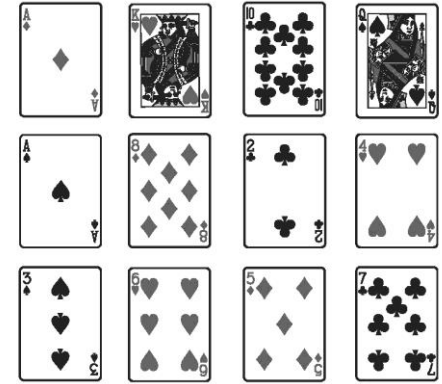
(e) $H(\text{card}) = 1.58$



(f) $H(\text{card}) = 3.58$

Figure 4.5: The entropy of different sets of playing cards measured in bits.

$$H(card) = -\sum_{i=1}^l P(card = i) \times \log_2(P(card = i))$$



$$(a) H(card) = -1 \times \log_2 1 = 0 \text{ bits}$$

$$(b) H(card) = -\left(\frac{3}{4} \times \log_2 \frac{3}{4} + \frac{1}{4} \times \log_2 \frac{1}{4}\right) = 0.81 \text{ bits}$$

$$(c) H(card) = -\left(\frac{1}{2} \times \log_2 \frac{1}{2} + \frac{1}{2} \times \log_2 \frac{1}{2}\right) = 1.00 \text{ bits}$$

$$(d) H(card) = -\left(\frac{1}{2} \times \log_2 \frac{1}{2} + \frac{1}{4} \times \log_2 \frac{1}{4} + \frac{1}{4} \times \log_2 \frac{1}{4}\right) = 1.50 \text{ bits}$$

$$(e) H(card) = -\left(\frac{1}{3} \times \log_2 \frac{1}{3} + \frac{1}{3} \times \log_2 \frac{1}{3} + \frac{1}{3} \times \log_2 \frac{1}{3}\right) = 1.58 \text{ bits}$$

$$(f) H(card) = -\left[\sum_{i=1}^{12} \frac{1}{12} \times \log_2 \frac{1}{12}\right] = 3.58 \text{ bits}$$

2. SHANNON'S ENTROPY MODEL

- **Table:** The relationship between the entropy of a message and the set it was selected from.

Entropy of a Message	Properties of the Message Set
High	A large set of equally likely messages.
Medium	A large set of messages, some more likely than others.
Medium	A small set of equally likely messages.
Low	A small set of messages with one very likely message.

CONTENTS

- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**

3. INFORMATION GAIN

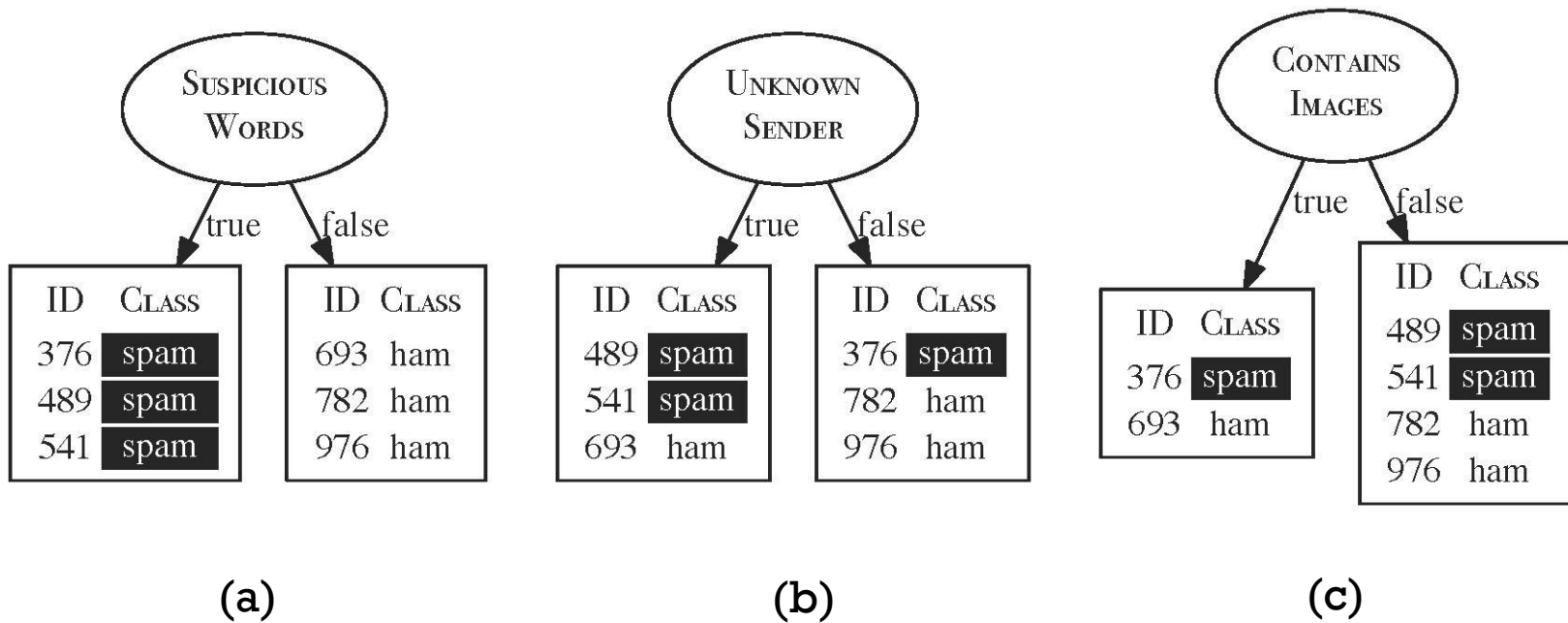
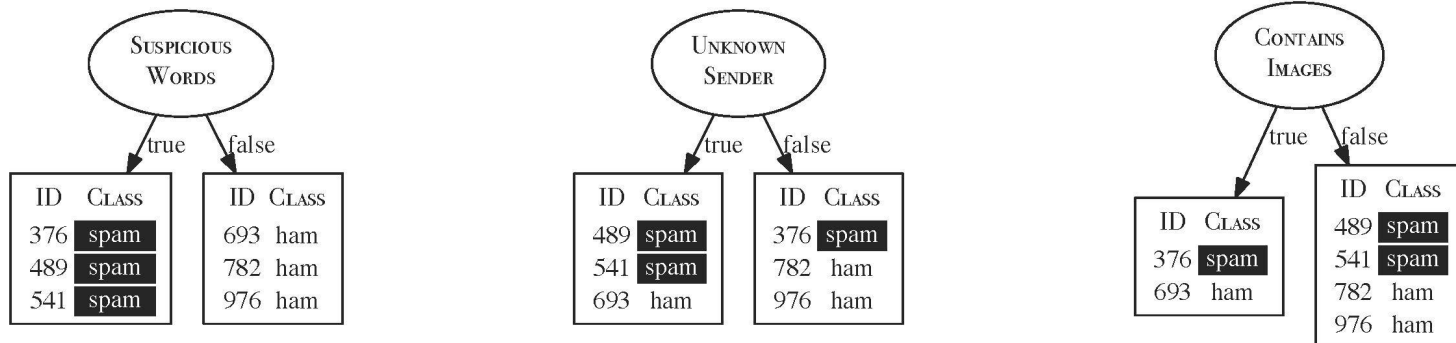


Figure 4.7: How the instances in the spam dataset split when we partition using each of the different descriptive features from the spam dataset in Table 4.2

3. INFORMATION GAIN

- Our **intuition** is that the ideal discriminatory feature will partition the data into **pure subsets** where all the instances in each subset have the same classification
 - SUSPICIOUS WORDS perfect split
 - UNKNOWN SENDER mixture but some information (when 'true' most instances are 'spam')
 - CONTAINS IMAGES no information
- One way to implement this idea is to use a metric called **information gain**



3. INFORMATION GAIN

Information Gain

- The information gain of a descriptive feature can be understood as a measure of **the reduction** in the overall entropy of a prediction task by testing on that feature.

3. INFORMATION GAIN

- Computing information gain involves the following 3 equations:

$$H(t, D) = - \sum_{l \in \text{levels}(t)} \left(P(t=l) \times \log_s(P(t=l)) \right) \quad (4.2)$$

$$\text{rem}(d, D) = \sum_{l \in \text{levels}(d)} \frac{|D_{d=l}|}{|D|} \times H(t, D_{d=l}) \quad (4.3)$$

weighting

entropy of
Partition $D_{d=l}$

$$IG(d, D) = H(t, D) - \text{rem}(d, D) \quad (4.4)$$

3. INFORMATION GAIN

- As an example we will calculate the information gain for each of the descriptive features in the spam email dataset.

3. INFORMATION GAIN

- Calculate the **entropy** for the target feature in the dataset.

$$H(t, D) = - \sum_{l \in \text{levels}(t)} \left(P(t = l) \times \log_s (P(t = l)) \right)$$

ID	SUSPICIOUS WORDS	UNKNOWN SENDER	CONTAINS IMAGES	CLASS
376	true	false	true	spam
489	true	true	false	spam
541	true	true	false	spam
693	false	true	true	ham
782	false	false	false	ham
976	false	false	false	ham

3. INFORMATION GAIN

$$\begin{aligned} H(t, D) &= - \sum_{l \in \{'spam', 'ham'\}} P(t = l) \times \log_s (P(t = l)) \\ &= - \left[P(t = 'spam') \times \log_2 (P(t = 'spam')) \right. \\ &\quad \left. + P(t = 'ham') \times \log_2 (P(t = 'ham')) \right] \\ &= - \left[\frac{3}{6} \times \log_2 \left(\frac{3}{6} \right) + \frac{3}{6} \times \log_2 \left(\frac{3}{6} \right) \right] \\ &= 1 \text{ bit} \end{aligned}$$

3. INFORMATION GAIN

- Calculate the **remainder** for the **SUSPICIOUS WORDS** feature in the dataset.

$$rem(d, D) = \sum_{l \in levels(d)} \frac{|D_{d=l}|}{|D|} \times H(t, D_{d=l})$$

ID	SUSPICIOUS WORDS	UNKNOWN SENDER	CONTAINS IMAGES	CLASS
376	true	false	true	spam
489	true	true	false	spam
541	true	true	false	spam
693	false	true	true	ham
782	false	false	false	ham
976	false	false	false	ham

3. INFORMATION GAIN

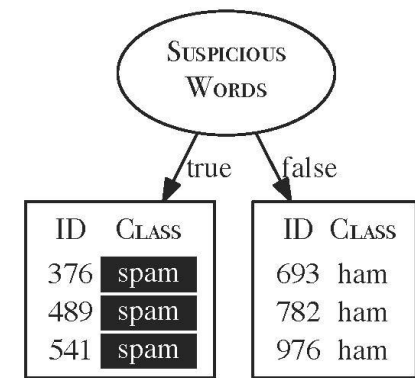
$rem(WORDS, D)$

$$= \left(\frac{|D_{WORDS=T}|}{|D|} \times H(t, D_{WORDS=T}) \right) + \left(\frac{|D_{WORDS=F}|}{|D|} \times H(t, D_{WORDS=F}) \right)$$

$$= \left\{ \frac{3}{6} \times \left[- \sum_{l \in \{ 'spam', 'ham' \}} P(t=l) \times \log_s(P(t=l)) \right] \right\} \\ + \left\{ \frac{3}{6} \times \left[- \sum_{l \in \{ 'spam', 'ham' \}} P(t=l) \times \log_s(P(t=l)) \right] \right\}$$

$$= \frac{3}{6} \times \left\{ - \left[\frac{3}{3} \times \log_2 \left(\frac{3}{3} \right) + \frac{0}{3} \times \log_2 \left(\frac{0}{3} \right) \right] \right\} \\ + \frac{3}{6} \times \left\{ - \left[\frac{0}{3} \times \log_2 \left(\frac{0}{3} \right) + \frac{3}{3} \times \log_2 \left(\frac{3}{3} \right) \right] \right\}$$

$= 0 \text{ bits}$



3. INFORMATION GAIN

- Calculate the **remainder** for the **UNKNOWN SENDER** feature in the dataset.

$$rem(d, D) = \sum_{l \in levels(d)} \frac{|D_{d=l}|}{|D|} \times H(t, D_{d=l})$$

ID	SUSPICIOUS WORDS	UNKNOWN SENDER	CONTAINS IMAGES	CLASS
376	true	false	true	spam
489	true	true	false	spam
541	true	true	false	spam
693	false	true	true	ham
782	false	false	false	ham
976	false	false	false	ham

3. INFORMATION GAIN

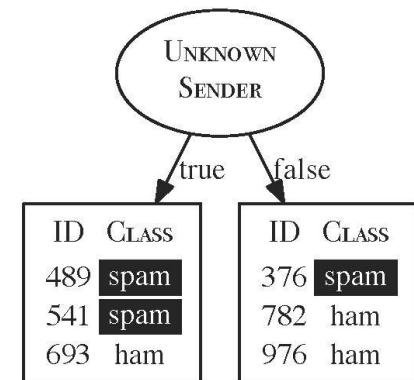
$rem(SENDER, D)$

$$= \left(\frac{|D_{SENDER=T}|}{|D|} \times H(t, D_{SENDER=T}) \right) + \left(\frac{|D_{SENDER=F}|}{|D|} \times H(t, D_{SENDER=F}) \right)$$

$$= \left\{ \frac{3}{6} \times \left[- \sum_{l \in \{'spam', 'ham'\}} P(t=l) \times \log_s(P(t=l)) \right] \right\} \\ + \left\{ \frac{3}{6} \times \left[- \sum_{l \in \{'spam', 'ham'\}} P(t=l) \times \log_s(P(t=l)) \right] \right\}$$

$$= \frac{3}{6} \times \left\{ - \left[\frac{2}{3} \times \log_2\left(\frac{2}{3}\right) + \frac{1}{3} \times \log_2\left(\frac{1}{3}\right) \right] \right\} \\ + \frac{3}{6} \times \left\{ - \left[\frac{1}{3} \times \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \times \log_2\left(\frac{2}{3}\right) \right] \right\}$$

$$= 0.9183 \text{ bits}$$



3. INFORMATION GAIN

- Calculate the remainder for the **CONTAINS IMAGES** feature in the dataset.

$$rem(d, D) = \sum_{l \in levels(d)} \frac{|D_{d=l}|}{|D|} \times H(t, D_{d=l})$$

ID	SUSPICIOUS WORDS	UNKNOWN SENDER	CONTAINS IMAGES	CLASS
376	true	false	true	spam
489	true	true	false	spam
541	true	true	false	spam
693	false	true	true	ham
782	false	false	false	ham
976	false	false	false	ham

3. INFORMATION GAIN

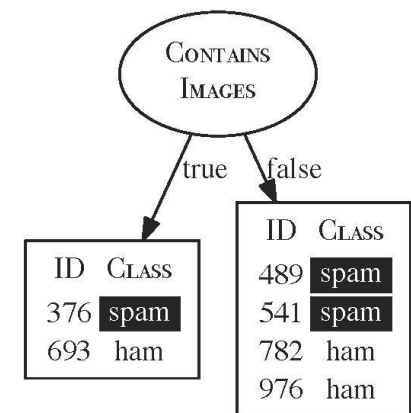
$rem(Images, D)$

$$= \left(\frac{|D_{Images=T}|}{|D|} \times H(t, D_{Images=T}) \right) + \left(\frac{|D_{Images=F}|}{|D|} \times H(t, D_{Images=F}) \right)$$

$$= \left\{ \frac{2}{6} \times \left[- \sum_{l \in \{ 'spam', 'ham' \}} P(t=l) \times \log_s(P(t=l)) \right] \right\} \\ + \left\{ \frac{4}{6} \times \left[- \sum_{l \in \{ 'spam', 'ham' \}} P(t=l) \times \log_s(P(t=l)) \right] \right\}$$

$$= \frac{2}{6} \times \left\{ - \left[\frac{1}{2} \times \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \times \log_2 \left(\frac{1}{2} \right) \right] \right\} \\ + \frac{4}{6} \times \left\{ - \left[\frac{2}{4} \times \log_2 \left(\frac{2}{4} \right) + \frac{2}{4} \times \log_2 \left(\frac{2}{4} \right) \right] \right\}$$

$= 1 \text{ bit}$



3. INFORMATION GAIN

- Calculate the **information gain** for the three descriptive feature in the dataset.

$$IG(d, D) = H(t, D) - \text{rem}(d, D)$$

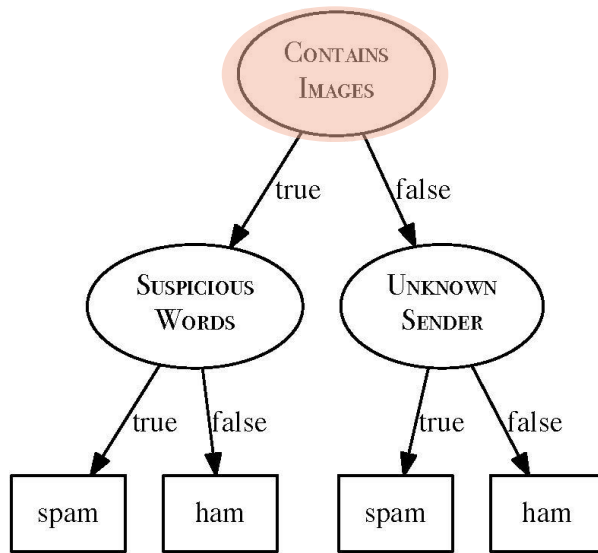
3. INFORMATION GAIN

- The results of these calculations match our intuitions.

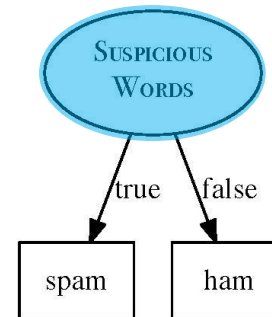
$$\begin{aligned} IG(\text{SUSPICIOUS WORDS}, D) \\ &= H(\text{CLASS}, D) - \text{rem}(\text{SUSPICIOUS WORDS}, D) \\ &= 1 - 0 = 1 \text{ bit} \end{aligned}$$

$$\begin{aligned} IG(\text{UNKNOWN SENDER}, D) \\ &= H(\text{CLASS}, D) - \text{rem}(\text{UNKNOWN SENDER}, D) \\ &= 1 - 0.9183 = 0.0817 \text{ bits} \end{aligned}$$

$$\begin{aligned} IG(\text{CONTAINS IMAGES}, D) \\ &= H(\text{CLASS}, D) - \text{rem}(\text{CONTAINS IMAGES}, D) \\ &= 1 - 1 = 0 \text{ bits} \end{aligned}$$



(a)



(b)

Figure 4.4: (a) and (b) show **two decision trees** that are consistent with the instances in the spam dataset.

CONTENTS

- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**

- ID3 Algorithm
 - Iterative Dichotomizer 3 (Quinlan, 1979)
- Attempts to create the **shallowest** tree that is consistent with the data that it is given.
- The ID3 algorithm builds the tree in a **recursive, depth-first** manner, beginning at the root node and working down to the leaf nodes.

1. The algorithm begins by choosing the **best descriptive feature** to test (i.e., the best question to ask first) using **information gain**.
2. A **root node** is then added to the tree and labelled with the selected test feature.
3. The training dataset is then **partitioned** using the test.
4. For each partition a **branch** is grown from the node.
5. The process is then **repeated** for each of these branches using the relevant partition of the training set in place of the full training set and with the selected test feature **excluded** from further testing.

P134 typo

Algorithm 4.1 Pseudocode description of the ID3 algorithm.

Require: set of descriptive features $\mathbf{d}$

Require: set of training instances $\mathcal{D}$

Lines 1–6 control when a new leaf node is created in the tree.

```

1: if all the instances in  $\mathcal{D}$  have the same target level  $C$  then
2:   return a decision tree consisting of a leaf node with label  $C$ 
3: else if  $\mathbf{d}$  is empty then
4:   return a decision tree consisting of a leaf node with the label of the
      majority target level in  $\mathcal{D}$ 
5: else if  $\mathcal{D}$  is empty then
6:   return a decision tree consisting of a leaf node with the label of the
      majority target level of the dataset of the immediate parent node

```

```

7: else
8:    $\mathbf{d}[\text{best}] \leftarrow \arg \max_{d \in \mathbf{d}} IG(d, \mathcal{D})$ 
9:   make a new node,  $Node_{\mathbf{d}[\text{best}]}$ , and label it with  $\mathbf{d}[\text{best}]$ 
10:  partition  $\mathcal{D}$  using  $\mathbf{d}[\text{best}]$ 
11:  remove  $\mathbf{d}[\text{best}]$  from  $\mathbf{d}$ 
12:  for each partition  $\mathcal{D}_i$  of  $\mathcal{D}$  do
13:    grow a branch from  $Node_{\mathbf{d}[\text{best}]}$  to the decision tree created by rerun-
        ning ID3 with  $\mathcal{D} = \mathcal{D}_i$ 
14:  end for
15: end if

```

If none of these cases hold, the algorithm continues to recursively create interior nodes in Lines 7–13.

The algorithm defines **three situations** where the recursion stops and a leaf node is constructed:

1. All of the instances in the dataset have the same classification (target feature value) then return a leaf node tree with that classification as its label.
2. The set of features left to test is empty then return a leaf node tree with the **majority class** of the dataset as its classification.
3. The dataset is empty return a leaf node tree with the **majority class** of the dataset at the **parent node** that made the recursive call.

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

- **Table 4.3:** The vegetation classification dataset.

ID	STREAM	SLOPE	ELEVATION	VEGETATION	
1	false	steep	high	chaparral	灌木丛
2	true	moderate	low	riparian	河岸带
3	true	steep	medium	riparian	
4	false	steep	medium	chaparral	针叶树
5	false	flat	high	conifer	
6	true	steep	highest	conifer	
7	true	steep	high	chaparral	

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

$$H(\text{VEGETATION}, D)$$

$$= - \sum_{l \in \left\{ \begin{array}{l} \text{'chaparral'}, \\ \text{'riparian'}, \\ \text{'conifer'} \end{array} \right\}} P(\text{VEGETATION} = l) \times \log_s (P(\text{VEGETATION} = l))$$

$$= - \left[\frac{3}{7} \times \log_2 \left(\frac{3}{7} \right) + \frac{2}{7} \times \log_2 \left(\frac{2}{7} \right) + \frac{2}{7} \times \log_2 \left(\frac{2}{7} \right) \right]$$

$$= 1.5567 \text{ bits}$$

ID	STREAM	SLOPE	ELEVATION	VEGETATION
1	false	steep	high	chaparral
2	true	moderate	low	riparian
3	true	steep	medium	riparian
4	false	steep	medium	chaparral
5	false	flat	high	conifer
6	true	steep	highest	conifer
7	true	steep	high	chaparral

CONTRIBUTIONS

under (Rem.) and
the dataset in Table 4.3

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_1$	$\mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_6, \mathbf{d}_7$	1.5	1.2507	0.3060
	'false'	$\mathcal{D}_2$	$\mathbf{d}_1, \mathbf{d}_4, \mathbf{d}_5$	0.9183		
SLOPE	'flat'	$\mathcal{D}_3$	$\mathbf{d}_5$	0	0.9793	0.5774
	'moderate'	$\mathcal{D}_4$	$\mathbf{d}_2$	0		
	'steep'	$\mathcal{D}_5$	$\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6, \mathbf{d}_7$	1.3710		
ELEVATION	'low'	$\mathcal{D}_6$	$\mathbf{d}_2$	0	0.6793	0.8774
	'medium'	$\mathcal{D}_7$	$\mathbf{d}_3, \mathbf{d}_4$	1.0		
	'high'	$\mathcal{D}_8$	$\mathbf{d}_1, \mathbf{d}_5, \mathbf{d}_7$	0.9183		
	'highest'	$\mathcal{D}_9$	$\mathbf{d}_6$	0		

ID	STREAM	SLOPE	ELEVATION	VEGETATION
1	false	steep	high	chaparral
2	true	moderate	low	riparian
3	true	steep	medium	riparian
4	false	steep	medium	chaparral
5	false	flat	high	conifer
6	true	steep	highest	conifer
7	true	steep	high	chaparral

CONTRIBUTIONS

Order (Rem.) and
dataset in Table 4.3

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_1$	$\mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_6, \mathbf{d}_7$	1.5	1.2507	0.3060
	'false'	$\mathcal{D}_2$	$\mathbf{d}_1, \mathbf{d}_4, \mathbf{d}_5$	0.9183		
SLOPE	'flat'	$\mathcal{D}_3$	$\mathbf{d}_5$	0	0.9793	0.5774
	'moderate'	$\mathcal{D}_4$	$\mathbf{d}_2$	0		
	'steep'	$\mathcal{D}_5$	$\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6, \mathbf{d}_7$	1.3710		
ELEVATION	'low'	$\mathcal{D}_6$	$\mathbf{d}_2$	0	0.6793	0.8774
	'medium'	$\mathcal{D}_7$	$\mathbf{d}_3, \mathbf{d}_4$	1.0		
	'high'	$\mathcal{D}_8$	$\mathbf{d}_1, \mathbf{d}_5, \mathbf{d}_7$	0.9183		
	'highest'	$\mathcal{D}_9$	$\mathbf{d}_6$	0		

ID	STREAM	SLOPE	ELEVATION	VEGETATION
1	false	steep	high	chaparral
2	true	moderate	low	riparian
3	true	steep	medium	riparian
4	false	steep	medium	chaparral
5	false	flat	high	conifer
6	true	steep	highest	conifer
7	true	steep	high	chaparral

CONTRIBUTIONS

under (Rem.) and
the dataset in Table 4.3

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_1$	$\mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_6, \mathbf{d}_7$	1.5	1.2507	0.3060
	'false'	$\mathcal{D}_2$	$\mathbf{d}_1, \mathbf{d}_4, \mathbf{d}_5$	0.9183		
SLOPE	'flat'	$\mathcal{D}_3$	$\mathbf{d}_5$	0	0.9793	0.5774
	'moderate'	$\mathcal{D}_4$	$\mathbf{d}_2$	0		
	'steep'	$\mathcal{D}_5$	$\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6, \mathbf{d}_7$	1.3710		
ELEVATION	'low'	$\mathcal{D}_6$	$\mathbf{d}_2$	0	0.6793	0.8774
	'medium'	$\mathcal{D}_7$	$\mathbf{d}_3, \mathbf{d}_4$	1.0		
	'high'	$\mathcal{D}_8$	$\mathbf{d}_1, \mathbf{d}_5, \mathbf{d}_7$	0.9183		
	'highest'	$\mathcal{D}_9$	$\mathbf{d}_6$	0		

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

Table 4.4: Partition sets (Part.), entropy, remainder (Rem.) and information gain (Info. Gain) by feature for the dataset in Table 4.3

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_1$	$\mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_6, \mathbf{d}_7$	1.5	1.2507	0.3060
	'false'	$\mathcal{D}_2$	$\mathbf{d}_1, \mathbf{d}_4, \mathbf{d}_5$	0.9183		
SLOPE	'flat'	$\mathcal{D}_3$	$\mathbf{d}_5$	0	0.9793	0.5774
	'moderate'	$\mathcal{D}_4$	$\mathbf{d}_2$	0		
	'steep'	$\mathcal{D}_5$	$\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6, \mathbf{d}_7$	1.3710		
ELEVATION	'low'	$\mathcal{D}_6$	$\mathbf{d}_2$	0	0.6793	0.8774
	'medium'	$\mathcal{D}_7$	$\mathbf{d}_3, \mathbf{d}_4$	1.0		
	'high'	$\mathcal{D}_8$	$\mathbf{d}_1, \mathbf{d}_5, \mathbf{d}_7$	0.9183		
	'highest'	$\mathcal{D}_9$	$\mathbf{d}_6$	0		

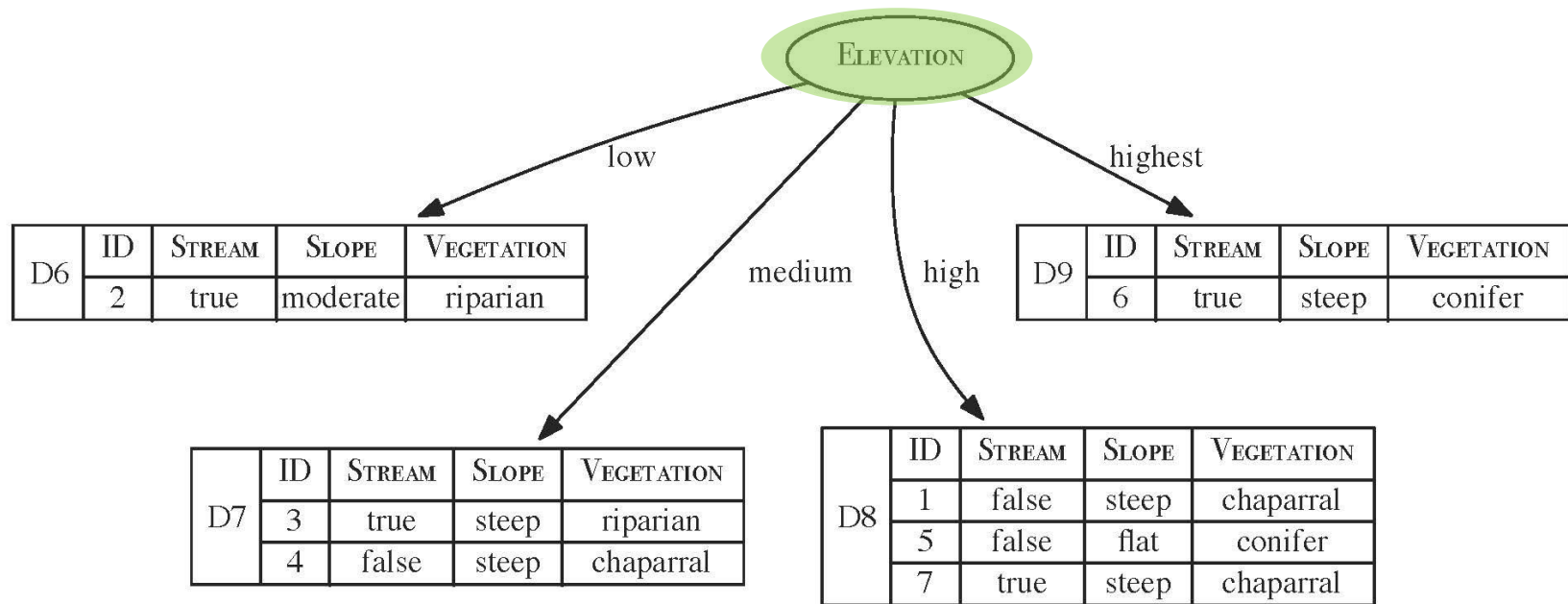


Figure 4.8: The decision tree after the data has been split using **ELEVATION**.

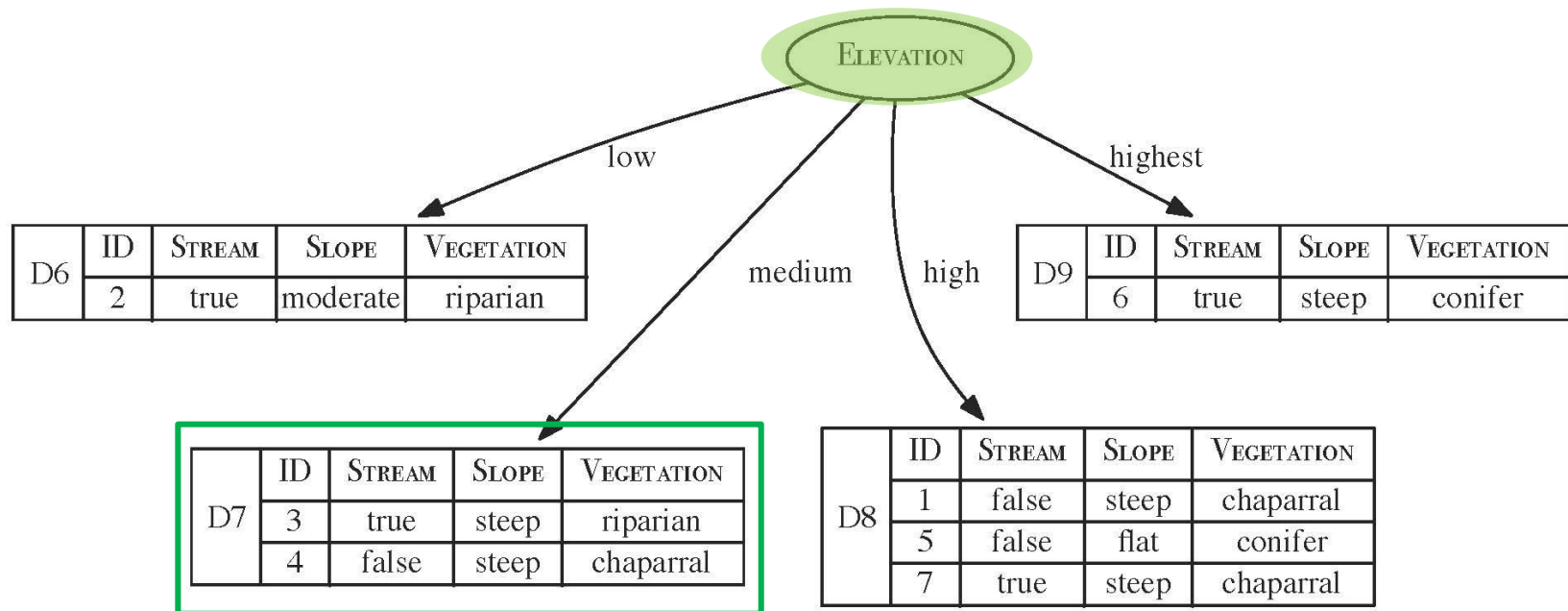


Figure 4.8: The decision tree after the data has been split using **ELEVATION**.

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

$$H(\text{VEGETATION}, D_7)$$

$$= - \sum_{l \in \left\{ \begin{array}{l} \text{'chaparral'}, \\ \text{'riparian'}, \\ \text{'conifer'} \end{array} \right\}} P(\text{VEGETATION} = l) \times \log_s (P(\text{VEGETATION} = l))$$

$$= - \left[\frac{1}{2} \times \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \times \log_2 \left(\frac{1}{2} \right) + \frac{0}{2} \times \log_2 \left(\frac{0}{2} \right) \right]$$
$$= 1.0 \text{ bit}$$

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

- **Table 4.5:** Partition sets (Part.), entropy, remainder (Rem.) and information gain (Info. Gain) by feature for the **dataset D7** in Figure 4.8.

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_{10}$	$\mathbf{d}_3$	0	0	1.0
	'false'	$\mathcal{D}_{11}$	$\mathbf{d}_4$	0		
	'flat'	$\mathcal{D}_{12}$		0		
SLOPE	'moderate'	$\mathcal{D}_{13}$		0	1.0	0
	'steep'	$\mathcal{D}_{14}$	$\mathbf{d}_3, \mathbf{d}_4$	1.0		

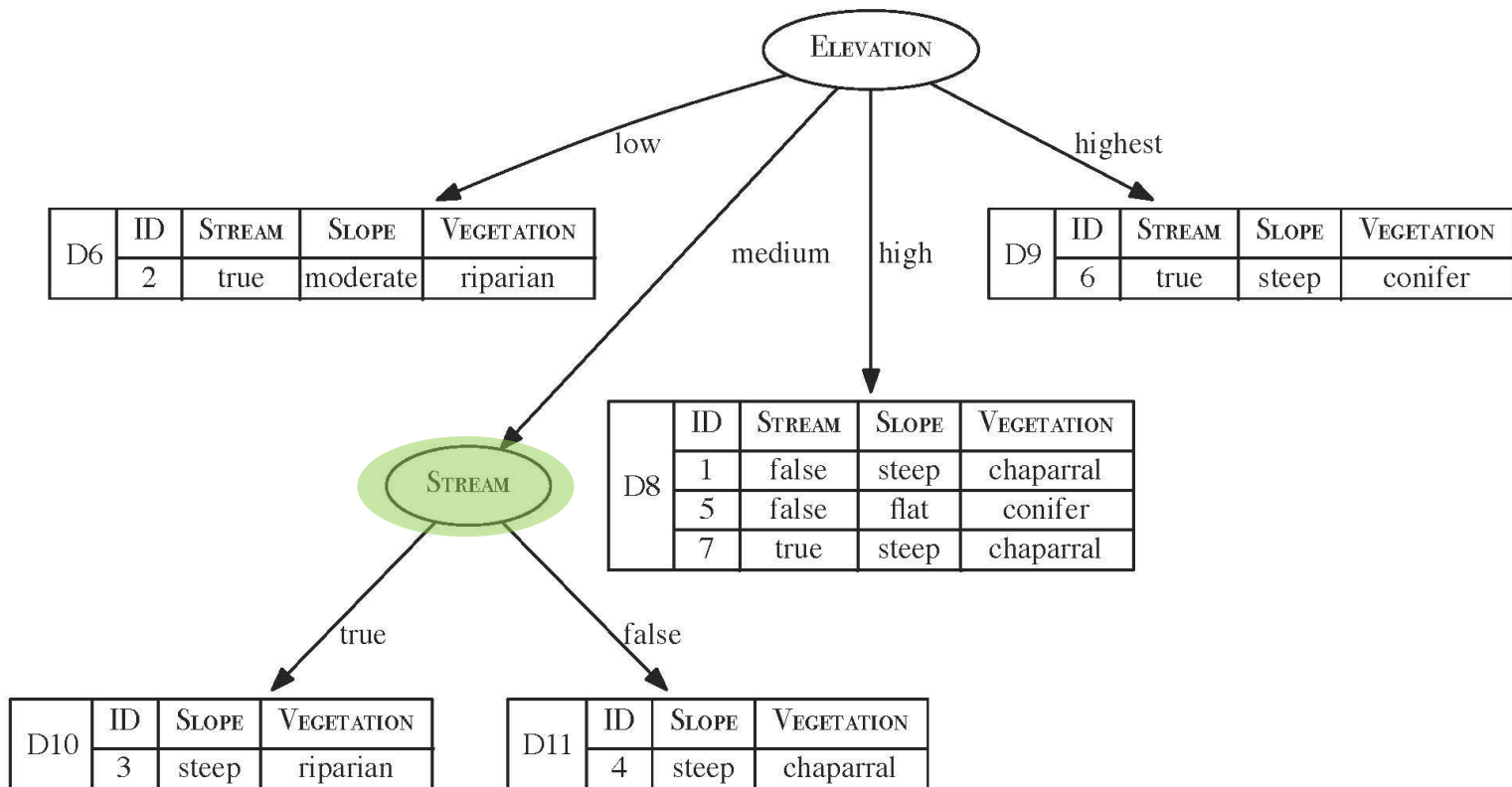


Figure 4.9: The state of the decision tree after the D7 partition has been split using STREAM.

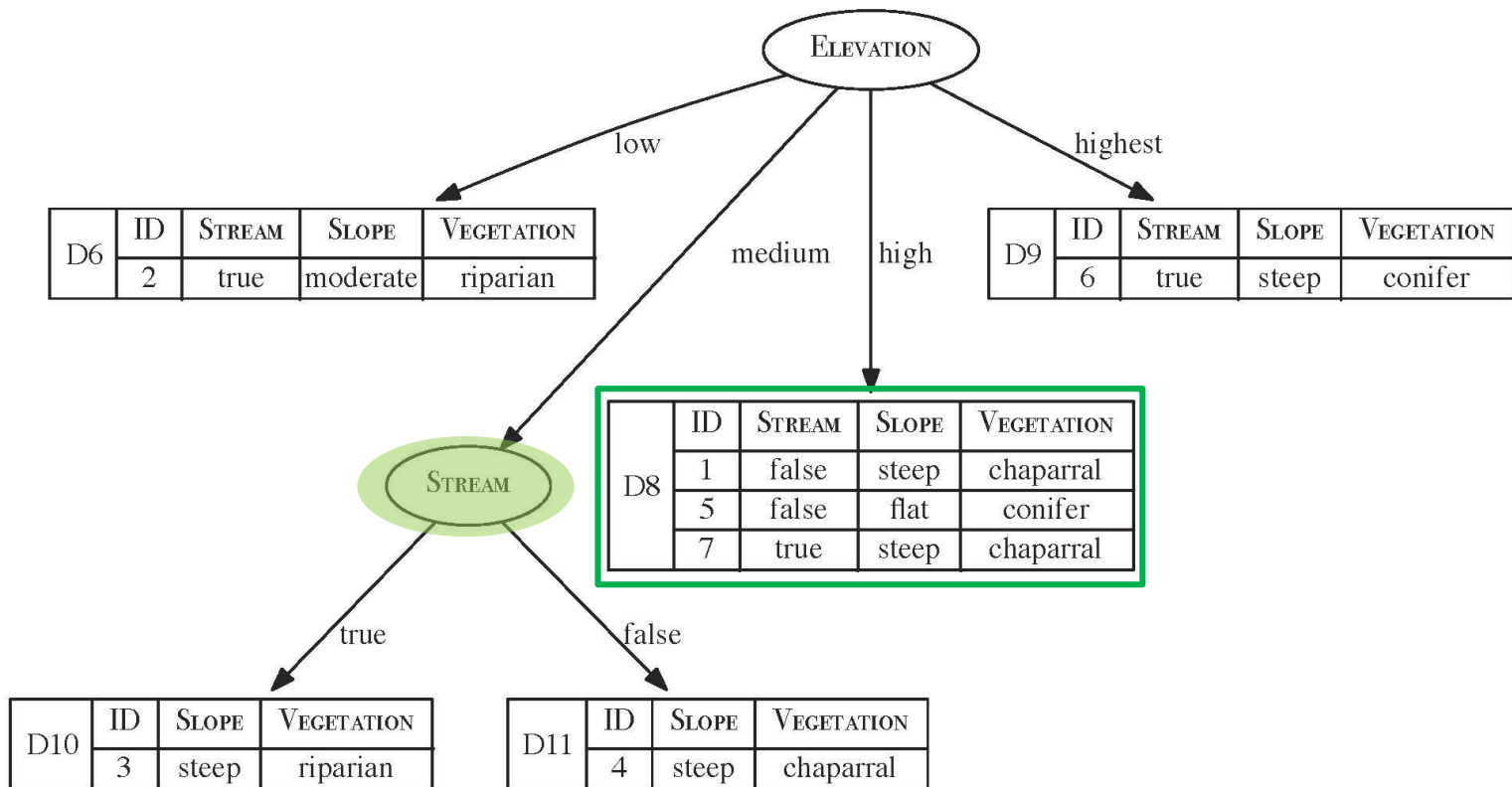


Figure 4.9: The state of the decision tree after the D7 partition has been split using STREAM.

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

$$H(\text{VEGETATION}, D_8)$$

$$= - \sum_{l \in \left\{ \begin{array}{l} \text{'chaparral'}, \\ \text{'riparian'}, \\ \text{'conifer'} \end{array} \right\}} P(\text{VEGETATION} = l) \times \log_s (P(\text{VEGETATION} = l))$$

$$= - \left(\left(\frac{2}{3} \times \log_2 \left(\frac{2}{3} \right) \right) + \left(\frac{0}{3} \times \log_2 \left(\frac{0}{3} \right) \right) + \left(\frac{1}{3} \times \log_2 \left(\frac{1}{3} \right) \right) \right)$$

$$= 0.9183 \text{ bits}$$

A WORKED EXAMPLE: PREDICTING VEGETATION DISTRIBUTIONS

- **Table 4.6:** Partition sets (Part.), entropy, remainder (Rem.) and information gain (Info. Gain) by by feature for the [dataset D8](#) in Figure 4.9.

Split By Feature	Level	Part.	Instances	Partition Entropy	Rem.	Info. Gain
STREAM	'true'	$\mathcal{D}_{15}$	$\mathbf{d}_7$	0	0.6666	0.2517
	'false'	$\mathcal{D}_{16}$	$\mathbf{d}_1, \mathbf{d}_5$	1.0		
SLOPE	'flat'	$\mathcal{D}_{17}$	$\mathbf{d}_5$	0	0	0.9183
	'moderate'	$\mathcal{D}_{18}$		0		
	'steep'	$\mathcal{D}_{19}$	$\mathbf{d}_1, \mathbf{d}_7$	0		

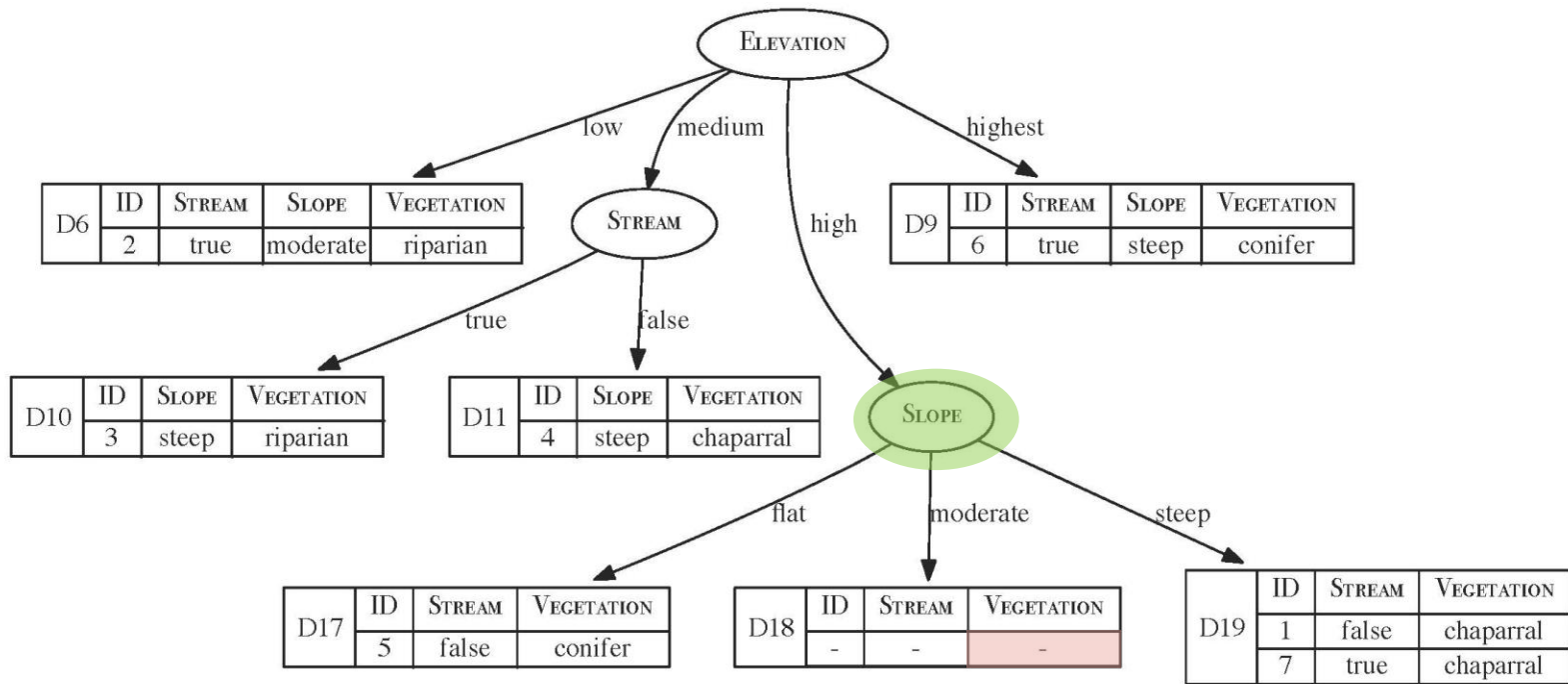


Figure 4.10: The state of the decision tree after the D8 partition has been split using SLOPE.

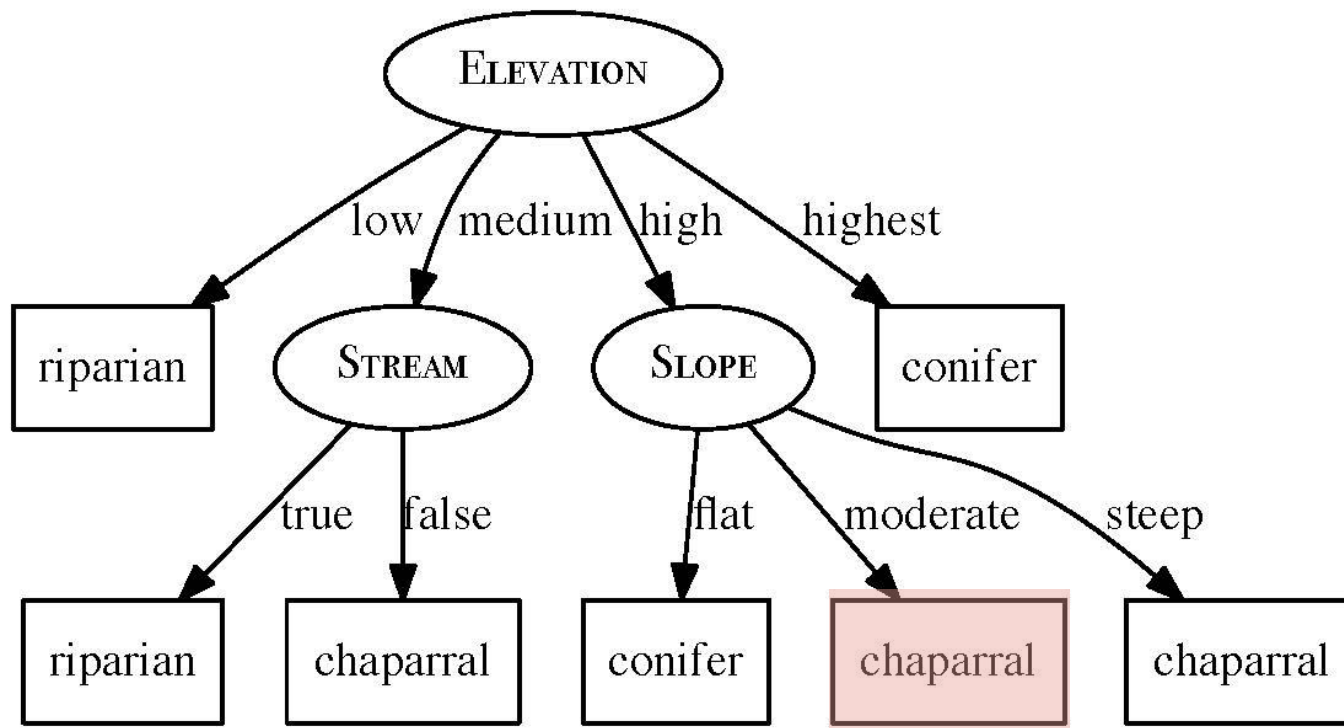
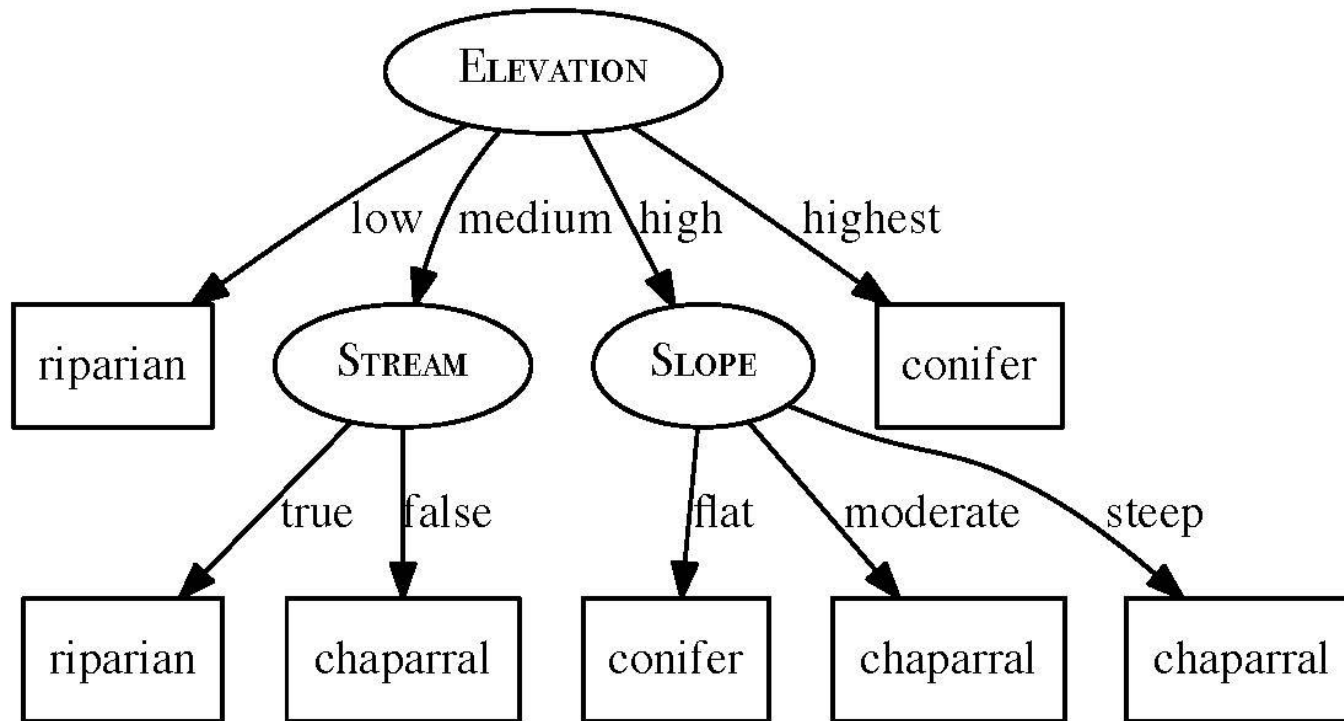
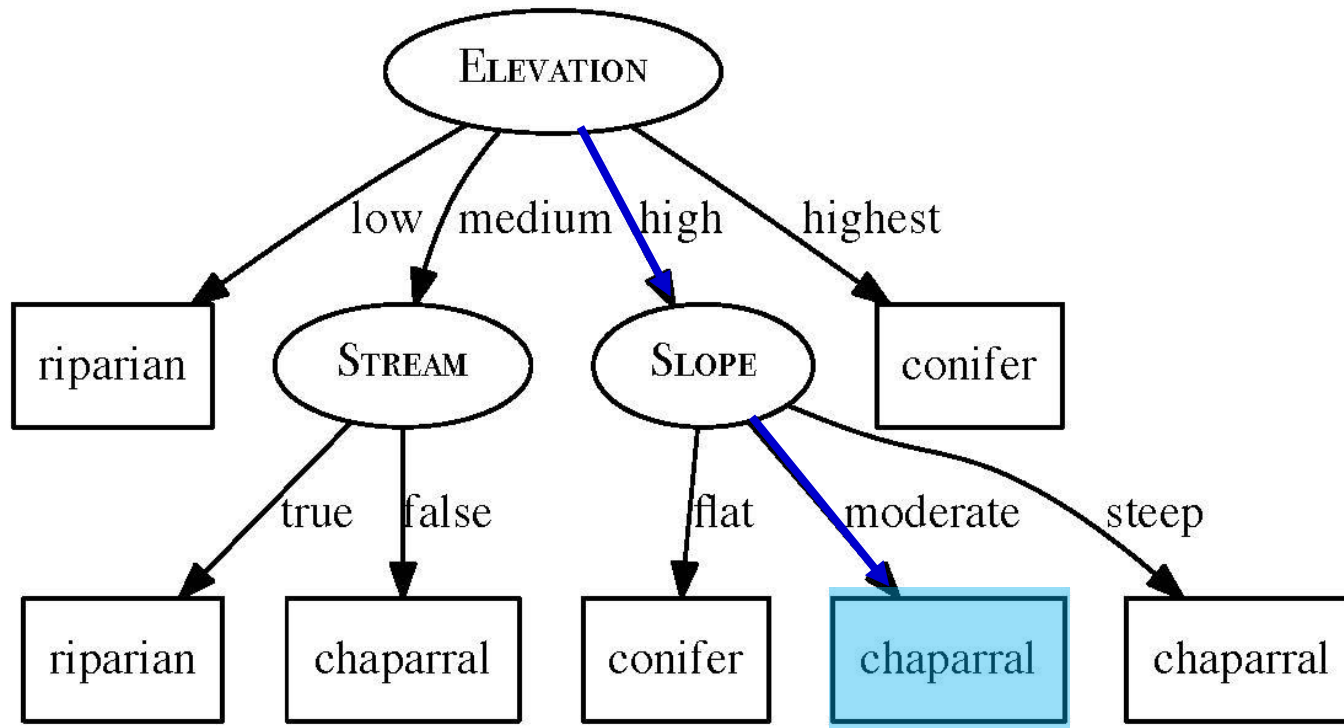


Figure 4.11: The final vegetation classification decision tree.



- What prediction will this decision tree model return for the following query?

STREAM = *'true'*, SLOPE=*'Moderate'*, ELEVATION=*'High'*



- What prediction will this decision tree model return for the following query?

STREAM = *'true'*, SLOPE=*'Moderate'*, ELEVATION=*'High'*

VEGETATION = *'Chaparral'*

ID	STREAM	SLOPE	ELEVATION	VEGETATION
1	false	steep	high	chaparral
2	true	moderate	low	riparian
3	true	steep	medium	riparian
4	false	steep	medium	chaparral
5	false	flat	high	conifer
6	true	steep	highest	conifer
7	true	steep	high	chaparral

STREAM = 'true', SLOPE='Moderate', ELEVATION='High'

VEGETATION = 'Chaparral'

- This is an example where the model is attempting to generalize **beyond** the dataset.
- Whether or not the generalization is **correct** depends on whether the assumptions used in generating the model (i.e. the inductive bias) were **appropriate**.

CONTENTS

- **Big Idea**
- **Fundamentals**
 - Decision Trees
 - Shannon's Entropy Model
 - Information Gain
- **Standard Approach: The ID3 Algorithm**
 - A Worked Example: Predicting Vegetation Distributions
- **Summary**

- The ID3 algorithm works the same way for **larger** more **complicated** datasets.
- There have been many **extensions** and **variations** proposed for the ID3 algorithm:
 - using different impurity measures (Gini, Gain Ratio)
 - handling **continuous** descriptive features
 - handling **continuous** targets
 - **pruning** to guard against over-fitting
 - using decision trees as part of an **ensemble** (Random Forests)
- We cover these extensions in Section 4.4.

HOMEWORK

- Reading: Section 4.1~4.3